10D — JianghuSect 2 (JS2-v2.6.0) Deep-Dive Feature Audit: Economy, Supply Chains, CP & External Trade (Part 4 of 5: Visiting Merchants, Instant Gear Liquidation Bug, Zero Member Bargaining, Spawned-Item Taint & Market Multipliers)

Target Codebase: JS2-v2.6.0 (js/external_market.js, js/economy.js, js/cpeco.js, js/sect.js, js/content.js, test/external_market_expansion_suite.js) **Target Audience**: Lead AI Systems Architects & Autonomous Coding Agents (**GPT** & **Gemini**) **Core Game Design Philosophy Mandate**:

1. **Closed-Loop Inter-Entity Trade**: Visiting merchants must draw starting capital from and return unspent/earned Spirit Stones to W.world.resourcePool.stones, and trade strictly within their finite v.stones and v.wants demand.
2. **Skill-Driven Bargaining & Path Identity**: A disciple's trading skill, socialCommerce aptitude, MERCANTILE_DAO subpath, and equipped ABACUS must genuinely affect prices—never silently hardcoded away.
3. **Strict Anti-Minting Hygiene (EC-06)**: Player-spawned materials (spawnedMats) must never be laundered through crafting into clean Spirit-Stone-generating pills or equipment.

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1. Executive Architecture Overview (Group 4 Scope)

js/external_market.js (g.EXT) bridges the internal Contribution-Point (CP) economy with the outside Spirit-Stone economy via 4 visiting trader archetypes (general, medicine, caravan, rare), autonomous sect treasury purchasing/surplus selling (EXT.tradeSect), stale equipment liquidation (EXT.sellStaleSectGear), and autonomous personal disciple trading (EXT.memberQuote / EXT.tradeMember / AI.ACTS.outsideTrade).

Cross-tracing external_market.js line-by-line against economy.js, cpeco.js, and sect.js exposes **five severe systemic bugs**:

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2. [ECO-EXT-01] Instant 0-Millisecond Buy-Then-Dump Liquidation Bug (external_market.js:285 vs 339 & 358 — -35% Treasury Burn!)

2.1 Exact Code Defect

1. In `EXT.tradeSect(W, v)` (`lines 285–288`), when the Sect Treasury buys needed authored equipment (`'EQ:' + eqId`) from a visiting merchant for its disciples (`priority = 15 + upgraders`, unit price `~1.12x base`):

```javascript
for (let gi = 0; gi < qty; gi++) W.sect.gear.push(ECO.mintGear(it, 1.0));
```

   - Notice two errors on that single line: a) `W` is omitted from `ECO.mintGear(it, 1.0)`. b) **`listedAtDay` is NEVER stamped on the newly purchased equipment** (`eq.listedAtDay === undefined`)!
2. Now trace **50 lines further down inside the EXACT SAME `EXT.tradeSect(W, v)` execution** (`line 339` -> `EXT.sellStaleSectGear`, `lines 358–359`):

```javascript
const listedAt = eq.listedAtDay != null ? eq.listedAtDay : 0;
const ageDays = (W.day || 0) - listedAt;
if (ageDays < graceDays) continue;
```

   - Because `W.day` starts at `360+` (Year 2+) or any day `>= 6`, `ageDays = W.day - 0 >= 6` evaluates **`true` IMMEDIATELY**!
   - Remember `boughtKeys` only tracks `'EQ:' + eqId` for material/pill loops (`lines 299–303`)—**`EXT.sellStaleSectGear` NEVER checks `boughtKeys`!**
   - Consequently, in the **exact same millisecond**, the Sect Treasury buys an upgrade sword from the merchant for `112` Spirit Stones, pushes it into `W.sect.gear` with `listedAtDay = undefined`, immediately misidentifies it as *"unsold stale stock older than 6 days"*, **sells it right back to the exact same merchant for `75` Spirit Stones (`-33%` net loss!), and permanently deletes the weapon from the simulation** before any disciple can walk to the shop to buy it!
   - The exact same instant-deletion bug hits **experimental prototype gear** minted in `content.js` and **underwater chest gear** minted in `fishing.js`!

2.2 Architectural Fix

- In `ECO.mintGear(W, r, q)` (`js/economy.js:244`), stamp `listedAtDay: (W && Number.isFinite(W.day)) ? W.day : 0` by default whenever `W` is provided, and pass `W` in `external_market.js:287` (`const gObj = ECO.mintGear(W, it, 1.0); gObj.listedAtDay = W.day;`).
- In `EXT.sellStaleSectGear`, skip any item whose `'EQ:' + eq.id` is in `opts.boughtKeys`, and if `eq.listedAtDay == null`, initialize `eq.listedAtDay = W.day` and wait the full `graceDays` before liquidating!

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3. `[ECO-EXT-02]` Disciple Bargaining Skill Is 100% Dead in `EXT.quotePrice` (`external_market.js:76–82` vs `445`)!

3.1 Exact Code Defect

- Look at `AI.ACTS.outsideTrade.score` (`line 445`):

```
JavaScript
S.add('ต่อรองเป็น', (n.h.skills.trading || 0) / 350);
```

> The AI tells the player that the disciple chose to trade with the merchant *"because they know how to bargain (ต่อรองเป็น)"*!

- Now look at `EXT.quotePrice(W, id, side, skill)` (`lines 76–82`):

```
JavaScript
if (side === 'sect-buy') mult = 1.15 - skill / 850;
else if (side === 'sect-sell') mult = 0.78 + skill / 900;
else if (side === 'member-buy') mult = 1.18;
else if (side === 'member-sell') mult = 0.76;
```

> And in `EXT.memberQuote` (`lines 393, 405, 412, 418, 428, 434`), **`skill` isn't even passed (`EXT.quotePrice(W, id, 'member-buy')`)**!

  - Whether a disciple is an illiterate brute (`trading = 0`) or a Stage-8 **Grandmaster of `MERCANTILE_DAO`** wielding the **Legendary Golden Abacus of Caishen** (`trading = 100`, `socialCommerce = 98`, `persuasion = 95`),

both pay the exact flat `1.18x` markup and sell at the exact flat `0.76x` markdown (`0.00%` bargaining effect)!

- Furthermore, `v.gearStock` procedural pricing (`line 422`: `Math.round((gr.value || 35) * 1.15)`) bypasses `EXT.quotePrice`, `W.market.mult`, AND `FREE_TRADE_TREATY` (`-20%` tariff)!

3.2 Architectural Fix

- Compute `memberTradeSkill(n)`: `clamp((E(n, 'intellect.numeracy') * 0.3 + E(n, 'aptitudes.socialCommerce') * 0.3 + (n.h.skills.trading || 0) * 0.4) + (n.cult?.path === 'MERCANTILE_DAO' ? 15 : 0) + (M.equippedType(n) === 'ABACUS' ? 8 : 0), 0, 115)`.
- Apply `skill` in `EXT.quotePrice` for `'member-buy'` (`1.18 - skill / 750`) and `'member-sell'` (`0.76 + skill / 800`), and apply active `FREE_TRADE_TREATY` tariff discounts consistently across both `v.stock` and `v.gearStock`.

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4. `[ECO-EXT-03]` Player-Spawned Input Laundering (`EC-06`) & Unsolicited Recipe Liquidation (`EC-09`)

4.1 Spawned Material -> Crafted Pill/Gear Laundering (`EC-06`)

- While raw `spawnedMats` and `spawnedPills` are blocked from direct merchant sale (`lines 310, 426, 432`), **the moment an NPC uses `spawnedMats` inside `ECO.reserveInputs` -> `ECO.finishCraft` (or `ECO.personalCraft`), the resulting pills or equipment are added to `W.sect.stock` / `W.sect.gear` WITHOUT incrementing `W.sect.spawnedPills[key]` or setting `gear.playerSpawned = true`!**
- On the very next merchant visit, `EXT.tradeSect` sells those spawned-input pills/gears for **real Spirit Stones**, laundering God-mode spawned items into legitimate economy revenue and distorting the `v2.6.0` Periodic Digest's separation between Account #2 (External Ledger) and Account #4 (Player Interventions)!
- **Fix**: In `ECO.reserveInputs`, record `ctx.usedSpawnedInput = true` whenever any consumed input unit decrements `W.sect.spawnedMats[matId]` or `n.inv.spawnedMats[matId]` (prioritizing natural stock first so spawned stock is only touched when natural stock is insufficient). If `ctx.usedSpawnedInput` is true, mark the crafted output as `playerSpawned = true` / `spawnedPills`!

4.2 Unsolicited Recipe Scroll Sale (`external_market.js:327–335`)

- Notice `surplusTechManuals` (`line 318`), `surplusSubManuals` (`line 323`), and `organs` (`line 336`) check `v.wants`, whereas **`surplusRecipes` (`lines 327–335`) sells duplicate recipe scrolls unconditionally WITHOUT checking `v.wants[key]`** (draining the merchant's `v.stones` on unwanted recipes before `sellStaleSectGear` runs)! Require `v.wants[key] > 0` or add recipe wants in `buildWants`.

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5. `[ECO-EXT-04]` Member-Bought Manual Double-Taxation & Merchant Pool Leak

1. **Personal Manual Confiscation (`external_market.js:465–470`)**:
   - In `EXT.tradeMember` (`fresh.side === 'buy-manual'`), the disciple pays **their own personal `n.inv.stones`**, yet lines `468–469` push `tid` into **BOTH `W.sect.library` AND `n.inv.techs`**.
   - At midnight, `CPE.daily` (`cpeco.js:487`) deletes `tid` from `n.inv.techs` because `W.sect.library.includes(tid)` is now true—forcing the disciple who already paid `100%` Spirit Stones out of pocket to **pay full `CP` tuition AGAIN** to read the book they bought!
   - **Fix**: Either keep `buy-manual` strictly in `n.inv.techs` (letting loyal disciples donate it later for `CPE.techReward` + a free personal `studyWaiver`, while selfish disciples study their private scroll directly), OR grant `n.ext.studyWaivers[tid] = true` immediately upon library deposit!
2. **Merchant `v.stones` Closed-Loop Conservation (`external_market.js:186 & 505`)**:
   - Fund `visitor.stones` from `W.world.resourcePool.stones` on arrival (`EXT.openVisit`) and return `visitor.stones` back to `W.world.resourcePool.stones` when the merchant departs (`EXT.daily`), preventing multi-decade drainage of the continental Spirit-Stone pool!
3. **Market Master (`marketmaster`) Synergy**:
   - Currently `treasurySkill(W)` (`line 214`) checks `SECT.holder(W, 'treasury')` only, completely ignoring an appointed **`marketmaster`** (ผู้ดูแลตลาดสำนัก)! Blend `Math.max(treasuryHolderSkill, marketMasterSkill)` and allow an honest `marketmaster` to attract `+1` extra merchant item slot per visit.